

WASSWA WILSON

Data Analyst · Biomedical Engineer · Full-Stack Software Developer · Researcher and Proposal Writer

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English (fluent, professional working proficiency) · Luganda (native)

PROFESSIONAL PROFILE

I am a biomedical programmes leader, data analyst and full-stack software developer based in Kampala, with over nine years of work across health research, data systems and software engineering. My biomedical work covers equipment inventory, needs assessment, planning and technical advisory. As Biomedical Programs Manager at the Gould Family Foundation (GFF), I led biomedical engineers and technicians across seven countries in East, Central and Southern Africa. I also trained doctors, nurses and apprentices alongside these teams. I founded and develop HOSSPI, my hospital management platform. My data work includes analysis in Python, SQL, MATLAB and Excel, data validation, automated auditing and management dashboards. At International Hospital Kampala, I managed the biomedical function, developed COVID-19 oxygen demand models for hospital and national planning, and supported COHSASA accreditation. My research experience includes laboratory data analysis and mathematical modelling at the Uganda Virus Research Institute. I also undertake independent biomedical consulting across five countries, develop software and production AI agents, and write institutional proposals.

CAREER HIGHLIGHTS

- As Biomedical Programs Manager at the Gould Family Foundation (GFF), I led biomedical engineers and technicians across Uganda, Kenya, Tanzania, Burundi, Malawi, Somaliland and Rwanda. I managed programme planning, implementation and reporting across partner health facilities, and I led training for biomedical engineers, technicians and clinical users, including doctors, nurses and apprentices.
- I founded and develop HOSSPI, a hospital management system covering patient records, clinical, laboratory, pharmacy and billing workflows. The platform includes biomedical engineering, role-based permissions, multi-facility management and live analytics.
- My current data analysis work includes community screening, referral and follow-up data in Python and SQL. I define reporting indicators and data-quality checks, and I develop FCHIP management dashboards alongside the platform's data capture, database and cloud flows.
- At International Hospital Kampala, I led upgrades to the theatre, laboratory, neonatal intensive care unit (NICU) and oxygen plant, and the installation of a 64-slice CT scanner and digital X-ray. I developed oxygen demand models for hospital and Ministry of Health planning, and supported successful COHSASA accreditation.
- I am building a cross-sector data collection and reporting application that writes the report as the user works, instead of leaving it to be compiled afterwards. I designed it for accounting, research, biology, agriculture, social media and similar fields.
- I designed, built and deployed an AI reception agent for a law firm in Texas, USA. It handles live inbound client calls, books and reschedules appointments, and runs client follow-up. It is in active production use.
- At GFF, I led the NICU upgrade at Mama Lucy Kibaki Hospital in Nairobi, Kenya, and developed standard operating procedures (SOPs) for supported equipment at partner facilities.

RESEARCH, DATA AND PROPOSAL DEVELOPMENT

Research

I worked as a research intern at the Uganda Virus Research Institute on laboratory data analysis, mathematical modelling and research pipeline design in Python. My BSc at Makerere covered Research Methods, Biomedical Modelling and Simulation, Biosignal Processing and Analysis, and Bioinformatics and Functional Genomics, and I carried a full research project through to completion. I apply the same method to operational problems, most visibly in the COVID-19 oxygen demand modelling.

Data management at scale

I built and maintained equipment registers, maintenance schedules and service histories at International Hospital Kampala and the Gould Family Foundation. I now design and run FCHIP's data layer, including capture schemas, validation rules, cloud flows, storage and synchronisation across web and mobile.

Analysis and reporting

I analyse data in Python, SQL, MATLAB and Excel, build live management dashboards, and develop automated auditing and verification tools that check records in bulk.

Proposals, grants and technical writing

I write and edit institutional proposals, grant applications, investment proposals, concept notes, partnership documents, pitch decks, technical documentation and user manuals, together with the annexes and evidence files that have to stand up to assessment.

Modern tooling and automation

I work with AI agents and agent-assisted development daily, in Cursor, Claude, OpenAI GPT and Codex, and I build automation pipelines that take routine collection, reconciliation and reporting off people's hands.

TECHNICAL SKILLS

Data, analytics and reporting

Python · SQL · MATLAB · Excel · relational modelling and query optimisation · data cleaning and validation · ETL and cloud data pipelines · dashboard and report design · automated auditing and verification tooling

AI, agents and automation

AI agent design, deployment and monitoring · workflow and process automation · third-party systems integration · Cursor · Claude · OpenAI GPT · Codex

Programming languages

Java · Python · JavaScript · TypeScript · C · C++ · C# · PHP · SQL · MATLAB · Arduino C · HTML5 · CSS3

Web and mobile

React · Next.js (App Router) · React Native · Expo · Tailwind CSS · styled-components · Bootstrap · JavaFX / FXML · Jest · Playwright

Backend and APIs

Node.js · Express · Laravel · CodeIgniter · Spring / Spring Boot · Convex · Prisma · MySQL · REST APIs · JWT · role-based access control

Cloud and DevOps

AWS · Microsoft Azure · Vercel · EAS · Bunny.net CDN · Git and GitHub · CI/CD pipelines · release and environment management

Research and scientific

Research methods and study documentation · mathematical modelling and simulation · biosignal processing · medical imaging · bioinformatics and functional genomics · embedded systems design · Proteus · Eagle CAD · Solid Edge

Documents and productivity

Microsoft Word · Excel · PowerPoint · proposal, grant and investment writing · technical writing · document editing and design · Windows, Linux and Android

Biomedical planning and advisory

Equipment inventory · equipment and needs assessment · facility master planning · technical advisory · lifecycle management · procurement specification · equipment SOP development

Medical equipment

CT and X-ray imaging · ultrasound and C-arms · ventilators, monitors and infusion pumps · theatre and ICU systems · haematology, chemistry and blood bank analysers · neonatal incubators, phototherapy, foetal Dopplers and CTG · oxygen plants, cylinder and manifold systems and piped medical gas · RO water and sterilisation plant

PROFESSIONAL EXPERIENCE

Founder and Developer

Current

HOSSPI Hospital Management System

- I founded HOSSPI and develop its hospital management platform, integrating patient records with clinical, laboratory, pharmacy, billing and biomedical engineering workflows.
- I develop role-based permissions, multi-facility management and live analytics to support hospital teams' operational and reporting needs.

Data Analyst and Lead Software Developer, FCHIP and Health Information Systems

2026 to present

FairBanks Medical Centre Limited, Kampala, Uganda

- I analyse community screening, referral and follow-up data in Python and SQL. I define reporting indicators, validation rules and data-quality checks, and I develop management dashboards based on verified figures.
- I lead the design and build of the FairBanks Community Health Intelligence Platform (FCHIP). I design the data capture forms, database structure and cloud data flows, and I deliver the platform across web and mobile.
- I write FairBanks proposals, including Community Health Reach, accelerator and fellowship applications, and maternity and diagnostics investments, with financial annexes and evidence. I prepare partnership documents and pitch decks. I chaired the meeting with the Uganda Health Partners Cooperative on cooperative health insurance in July 2026.
- I own the technical documentation, the data-protection practice, the access control and the release management across the platform, and I manage grant opportunities and deadlines.

Consultant Biomedical Engineer, Independent Practice

Mar 2025 to present

Contract assignments: Uganda, DR Congo, Kenya, Tanzania, Somaliland

- I install, commission, calibrate and maintain hospital medical equipment under contract across five countries, working independently on site and to a fixed handover standard.
- I completed the laboratory, theatre and radiology installations at several hospitals in the Democratic Republic of Congo, including the facility serving the Kibali Gold Mine.
- I train the clinical users at handover, and I produce a written service report and maintenance record for every assignment, so the client keeps a verifiable equipment history.

Biomedical Programs Manager

Aug 2024 to Feb 2025

Gould Family Foundation (GFF), East, Central and Southern Africa

- I led biomedical engineers and technicians across Uganda, Kenya, Tanzania, Burundi, Malawi, Somaliland and Rwanda. I managed programme planning, implementation and reporting across partner health facilities in all seven countries.
- I managed equipment inventories, lifecycle records, procurement documentation and programme reporting.
- I assessed equipment needs, advised on lifecycle management and master planning, and developed standard operating procedures (SOPs) for supported equipment at partner facilities.
- I led the upgrade of the neonatal intensive care unit (NICU) at Mama Lucy Kibaki Hospital in Nairobi, Kenya, using oxygen and equipment demand modelling to inform the capacity plan.
- I led training programmes for biomedical engineers, technicians and clinical users, including doctors, nurses and apprentices.

Biomedical Manager

Jan 2020 to Jan 2024

International Hospital Kampala (IHK)

- I managed the biomedical engineering function of a major private hospital for four years, including team supervision, workplans and reporting to executive management.
- I led oxygen therapy management through the COVID-19 response and I supported Ministry of Health oxygen planning. I developed the formulas used to calculate patient oxygen consumption, which gave managers a dependable basis for sizing plant capacity, cylinder stock and flow requirements rather than estimating them.
- I prepared and maintained the compliance evidence files reviewed for COHSASA accreditation, and I supported the hospital through successful accreditation.
- I led upgrades to the theatre, laboratory, neonatal intensive care unit (NICU) and oxygen plant. I led the installation of a 64-slice CT scanner and digital X-ray, commissioned equipment including ICU systems through to handover, and trained clinical users.
- I maintained equipment inventories, preventive maintenance schedules and service records that reduced downtime, and coordinated equipment assessment, procurement, technical specification and lifecycle planning.

Biomedical Engineer

Apr 2019 to Jan 2020

Norvik Hospital Ltd, Kampala

- I maintained and serviced the hospital diagnostic and patient-monitoring equipment, and I supported the installation and calibration of imaging and ICU equipment.
- I helped establish the preventive maintenance schedules and service follow-up records.

Research Intern

Dec 2018 to Apr 2019

Uganda Virus Research Institute (UVRI)

- I carried out laboratory data analysis, mathematical modelling and research pipeline design in Python.
- I contributed to biomedical research projects and laboratory experimentation, and I handled the recording, quality checking and documentation of research data so results could be reproduced.

Teaching Assistant

Mar 2016 to Aug 2017

Makerere University, College of Health Sciences

- I assisted in teaching biomedical engineering course units at undergraduate level.
- I supervised and supported the laboratory practical sessions for engineering students.

SELECTED PROJECTS

HOSSPI Hospital Management System

Current

Founder and Developer · Hospital software and data systems

I founded and develop HOSSPI, a hospital management system connecting patient records with clinical, laboratory, pharmacy and billing workflows. The platform integrates biomedical engineering, role-based access permissions, multi-facility management and live analytics, giving hospital teams a shared view of patient care, hospital operations and management reporting. Website: <https://hosspi.com/>

Mama Lucy Kibaki NICU upgrade and equipment SOPs

Aug 2024 to Feb 2025

Biomedical Programs Manager · Gould Family Foundation, Nairobi, Kenya and partner facilities

I led the NICU upgrade at Mama Lucy Kibaki Hospital in Nairobi, Kenya, with equipment and oxygen needs assessment informing capacity planning. I also developed SOPs for supported equipment at GFF partner facilities.

IHK clinical infrastructure upgrades and imaging installations

Jan 2020 to Jan 2024

Biomedical Manager · International Hospital Kampala, Uganda

I led upgrades to the theatre, laboratory, NICU and oxygen plant, and the installation of a 64-slice CT scanner and digital X-ray. My oxygen demand modelling supported hospital capacity decisions and Ministry of Health planning.

AI reception agent

2025 to present

Designer and Developer · Law firm, Texas, USA

I built and deployed an AI agent that answers live inbound client calls, books and reschedules appointments, and runs post-contact client follow-up. It is in active production use.

Shulekeeper school information system

2021 to 2024

Assistant Software Developer · Shulekeeper

I contributed front-end and back-end development, third-party integrations, testing, user training and documentation to a multi-user records and reporting platform built with React and Next.js, TypeScript, Laravel, MySQL and AWS.

Endoscopy image capture and reporting software

2017 to 2019

Lead Software Developer · St. Catherine Hospital and St. Francis Hospital

I led the build of a clinical image capture and procedure reporting system in JavaFX and MySQL. I did the algorithm design, the relational modelling, the legacy data migration, the on-site installation, the user training and the user manual.

Wekebere foetal heart-rate monitoring system

2017 to 2024

Electronics Developer and Programmer · Wekebere

I built an Android and Arduino monitoring system for third-trimester foetal heart-rate measurement, and I implemented the Java serial communication between device and phone, with Microsoft Azure cloud storage.

EDUCATION

Certificate in Leadership and Management in Health

2022

University of Washington

Grade A+. Planning, leadership and management of health programmes.

BSc Biomedical Engineering, Second Upper Honours

2012 to 2017

Makerere University, Kampala

Research-relevant course units: Research Methods, Biomedical Modelling and Simulation, Biosignal Processing and Analysis, Bioinformatics and Functional Genomics, Database Systems and Software Engineering. I completed a full research project as part of the degree.

Advanced Java Programming, Level 1

2015 to 2016

Green Bridge School of Open Technologies

Grade A+.

Uganda Advanced Certificate of Education (UACE)

2011

Mengo Secondary School

Principal subjects: Mathematics, Physics, Biology, Chemistry.

Uganda Certificate of Education (UCE)

2009

St. John's Wakiso Secondary School

Division 1.

TRAINING AND CERTIFICATION

- **COHSASA accreditation standards: in-service preparation and compliance.** International Hospital Kampala, 2020 to 2024.
- **Fire Safety, Prevention, Firefighting and Emergency Scene Management.** Fire Technologies Limited (IHK), 2021.

INTERESTS

I play music, the saxophone in particular. · I watch film. · I train and exercise regularly. · I enjoy travelling and working in unfamiliar places, which the regional field work has made second nature.

REFEREES

Eng. Richard Ssejongo

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Hospital Nsambya

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